

Host Analysis

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SUMMARY:

- Purpose:** Reconstruct what the attacker did on a compromised endpoint.
- When:** After triage confirms compromise on a specific machine.
- Core Approach:** Follow the chain — Connection → Process → Memory → Payload → File → Logs → Timeline → Root Cause.
- Key Steps:**
1. Recap triage (match netstat PIDs to processes).
 2. **Memory analysis** (fileless threats) — **Volatility** (pslist, pstree, cmdline, netscan).
 3. Decode payloads — **CyberChef** (Base64, UTF-16).
 4. Analyze initial access files — **oletools** (oleid, olevba for macros).
 5. Review logs — Event ID **4624** (logon), **4688** (process creation).
 6. Build timeline & check persistence — **Velociraptor** (MFT timeline), Autoruns.
 7. Correlate everything into attack story.

Main Tools: Volatility, CyberChef, oletools, Velociraptor, Event Viewer.

Why it matters: Reveals initial access, execution, persistence, and impact on the host.

When to perform: After initial triage confirms compromise on an endpoint.

Why: Reconstruct attacker actions on the machine — initial access, execution, persistence, impact, and root cause. Follow the evidence chain: Network connection → Process → Memory → Payload → File → Logs → Timeline.

- Key Tools & Purpose:**
- **Velociraptor:** Remote DFIR collection, queries, and timelines across endpoints.
 - **Volatility:** Memory forensics (especially for fileless malware that leaves little disk footprint).
 - **CyberChef:** Decode obfuscated/encoded payloads (Base64, UTF-16, etc.).
 - **oletools** (oleid, olevba): Analyze malicious Office documents and extract macros.
 - **Windows Event Viewer:** Review built-in logs for logins, processes, RDP, etc.

Investigation Workflow (Step-by-Step Order)

1. Quick triage recap (network + processes).
2. Memory analysis (fileless threats).
3. Decode payloads.
4. Analyze initial access files (e.g., documents).
5. Review logs.
6. Build timelines and check persistence.
7. Correlate findings to understand full attacker activity.

1. Quick Triage Recap

- **Tool:** Built-in Windows cmd
`netstat -ano`
and
cmd
`tasklist`
Why: Match suspicious network connections (PIDs) to running processes. Identifies what is communicating with attackers or other systems.

2. Memory Analysis (for fileless attacks)

- **Tool:** **Volatility** (run on a memory dump) Common commands:
Bash
`volatility -f memory.raw pslist`
or
Bash
`volatility -f memory.raw pstree`
Why: Lists processes and parent-child trees. Reveals hidden or injected processes that may not appear in live tasklist.
Bash
`volatility -f memory.raw cmdline`
Why: Shows full command-line arguments. Often reveals encoded PowerShell or suspicious parameters.
Bash
`volatility -f memory.raw netscan`
Why: Extracts network connections from memory. Useful when live netstat is insufficient.

3. Decode Obfuscated Payloads

- **Tool:** **CyberChef** (web or desktop) Recipes: Base64 decode → UTF-16LE decode. Or PowerShell one-liner:
PowerShell
`[System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String("ENCODED_STRING"))`
Why: Attackers heavily obfuscate payloads. Decoding reveals the real commands or next-stage malware.

4. File & Initial Access Analysis

- **Tool:** **oletools**
Bash
`oleid suspicious.doc`
Why: Detects macros or other malicious features in Office files.
Bash
`olevba suspicious.doc`
Why: Extracts and displays macro code (e.g., AutoOpen that runs PowerShell).

5. Log Analysis

- **Tool:** **Windows Event Viewer** (or tools like EvtxECmd/Chainsaw)
Focus on Security log:
 - Event ID **4624** (successful logon)
 - **4625** (failed logon)
 - **4688** (process creation) **Why:** Reveals logins (including RDP type 10), process executions, and privilege use. Correlate with triage findings.

6. Timeline & Persistence

- **Tool:** **Velociraptor** Use artifacts like Windows.Timeline.MFT or YARA scans. **Why:** Builds timelines of file activity, new executables, or downloads. Helps reconstruct sequence of events.
- **Tool:** **Autoruns** or registry checks (from triage) **Why:** Confirms persistence (run keys, services, tasks).

Mindset: Always follow the chain of evidence and correlate across tools to build a clear attack story on the host.